

# Lab 3: Architectural Pattern Selection – Written Justification

SRN: PES1UG24CS195 | Project: Attendance Management System

**Architecture Selection: We chose Layered Architecture for the Attendance Management System.**

## 1. Architectural Choice

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Layered Architecture, deployed as a web application (browser client, one application server). The **Presentation Layer** holds the Web UI with role-based dashboards for Administrator, Teacher and Student. The **Business Layer** holds Auth Manager (UC-01), Admin Manager (FR-07 to FR-09), Attendance Manager (UC-02 to UC-05), Percentage Calculator (UC-06, FR-11) and Report Manager (UC-07, FR-12). The **Data Layer** holds the Attendance Database. The Security / Auth Service and the Report Engine are external components reached only through the IVerifyCredentials and IReportEngine interfaces.

## 2. Two Reasons for the Choice

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- **Attendance data must stay accurate and never be duplicated.** FR-03 forbids two records for the same student, subject and date, and FR-04 and FR-11 derive percentages and low-attendance alerts from those records. With all attendance logic in one Attendance Manager writing to a single database, one transaction with a unique (student, subject, date) rule guarantees this and supports NFR-03 (no data loss). Microservices would spread the records and the percentage logic across separate services and databases, so the two could disagree.
- **The requirements ask for modular parts, and the scale is a single institution.** NFR-06 asks that features can be changed with minimal impact on others. Separate Auth, Admin, Attendance, Percentage and Report components behind interfaces give exactly that (for example, the Report Engine or the login service can be replaced without touching Attendance Manager). The system has 7 use cases, 3 user roles and one team, so the deployment and network cost of microservices is not justified. Plain client-server would leave all logic in one block and one server as a single point of failure for daily attendance marking.

## 3. Security Advantage

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NFR-01 requires that each role sees only what it is permitted to. In this design the Web UI never touches the database. Every request goes through a Business Layer interface, and Auth Manager applies role rules before any data is returned (a Student sees only their own records; only a Teacher can mark or update; only an Administrator manages users). Credentials are checked by the external Security / Auth Service, so passwords are not handled by attendance logic, and all client-server traffic uses HTTPS to protect data in transit.

## 4. Performance Benefit

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NFR-02 requires pages and reports within 3 seconds. Percentage Calculator gets its numbers through IRecordRead with aggregate SQL counts on indexed columns (student, subject, date), so it does not need to load every record into memory. Report compilation is handed to the Report Engine, so heavy reports do not delay teachers who are marking attendance. Because the layers are separate, the database can also be backed up on a schedule (NFR-03) and tuned without changing the user interface.

## Trade-off Acknowledged

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Layered designs add some call overhead between layers and scale as a whole rather than feature by feature. For an attendance system used by one institution, that cost is small compared with the accuracy, security and maintainability gained.